

Mathematical methods list of entry questions

1. Write down the Fourier series expression of a 2π -periodic function, $f(x)$, and express the coefficient of the $\cos 2x$ term as an integral.
2. Write down the complex Fourier transform $\hat{f}(\omega)$ of a function $f(x)$, and the inverse FT.
3. Express the Fourier transform of $f(x - b)$ with $\hat{f}(\omega)$ (derive the relation).
4. Define the convolution of two functions $f * g$ and state the basic properties of convolution.
5. How is the Dirac delta distribution defined? Calculate the Fourier transform of $\delta(x - 1)$.
6. Calculate the Fourier transform of

$$f(x) = \begin{cases} 1 & \text{if } -1 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$
7. Sketch the direction field of the ODE $y'(x) = y(x) + x$.
8. Derive the general solution of $y'(x) + p(x)y(x) = 0$.
9. Show that the ODE $y'(x) + p(x)y(x) = r(x)$ has a particular solution of the form

$$y_p(x) = e^{-h(x)} \int^x e^{h(\tilde{x})} r(\tilde{x}) d\tilde{x}, \quad \text{with } h(x) = \int^x p(\tilde{x}) d\tilde{x}.$$
10. Prove the superposition principle for the homogeneous linear ODE

$$y''(x) + a(x)y'(x) + b(x)y(x) = 0.$$
11. What is the general solution of the second-order linear homogeneous ODE with constant coefficients $2y''(x) - 2y'(x) - 12y(x) = 0$?
12. Consider the damped mass-spring system described by the ODE $y''(t) = -ky(t) - cy'(t)$. Analyze the exponential solution of this equation. What are the three types of motion, and in what parameter ranges do they occur?
13. How is the Wronskian defined? Using the Wronskian, determine if these two solutions are linearly independent: $y_1(x) = \cos 3x$, $y_2(x) = 4 \cos^3 x - 3 \cos x$.
14. Rewrite the fourth order ODE $y^{(4)}(x) + 2y'''(x) + 5y''(x) - 3y'(x) + y(x) = 0$ as a system of first order ODE's. How many initial conditions are needed?
15. Using separation of variables, solve the wave equation on the interval $[0, L]$ with Dirichlet boundary condition, a speed of sound c , and the initial condition $u(x, t = 0) = \sin(\pi x/L)$.
16. Using separation of variables, solve the heat equation on the interval $[0, L]$ with Dirichlet boundary condition, diffusion constant D , and initial condition $u(x, t = 0) = \sin(\pi x/L)$.
17. Find a solution to the heat equation $u_t = D u_{xx}$ in the form $u_0(x, t) = \frac{1}{\sqrt{t}} e^{-Ax^2/t}$ by finding A .
18. Write down the definition of e^z for $z = x + iy$ and prove that $e^{z_1} e^{z_2} = e^{z_1 + z_2}$.
19. Calculate $\text{Ln}(1 + i) - \text{Ln}(-i)$.
20. Find u and v in $f(x + iy) = u(x, y) + i v(x, y)$ for $f(z) = \cosh z$.
21. Write down the definition of complex differentiability at a point z_0 , and calculate the derivative of $f(z) = z^2$ from the definition.
22. Show that $f(z) = e^z$ is analytic, because u and v in $f(x + iy) = u(x, y) + i v(x, y)$ satisfy the Cauchy-Riemann equations. Show that u satisfies Laplace's equation.
23. Calculate the complex contour integral $\oint \frac{1}{z} dz$ where the contour runs around the unit circle counterclockwise.